Abhiram RS

Software engineer working across data pipelines and AI systems.

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Associate Software Engineer at EY GDS, keeping daily batch ETL pipelines running for an enterprise client, where debugging is the core of the job: following a failure across code, schedulers, logs and data files until the root cause is clear, then fixing it properly. Builds AI systems end to end on his own time. Open to roles in software engineering, data engineering and cloud operations.

Experience

Associate Software Engineer, EY GDS

Trivandrum | Nov 2025 to Present

- Responsible for keeping a suite of daily batch ETL pipelines (Python, Scala, PySpark, SQL) running reliably in the client's UAT environment, orchestrated through Tidal Enterprise Scheduler.
- Debugs failed loads end to end: traces each failure through scheduler output, Spark job behaviour and the source and target files on Oracle Cloud Infrastructure (browsed with CloudBerry), finds the root cause, fixes it and re-runs the load.
- Corrects historical data loads (internally called "history fixes") so previously processed periods are consistent with fixed logic and upstream data.
- Resolves defects in production-grade Scala and SQL transformation code alongside the wider delivery team.

Python, Scala, PySpark, SQL, Tidal Enterprise Scheduler, Oracle Cloud Infrastructure (OCI), CloudBerry

Full Stack Developer (Freelance, Remote), Perusal Gulf

Remote | Apr 2023 to Mar 2024

- Owned the backend and server hosting (another team member built the frontend) for a private, role-based welding-certification platform with four roles, enforcing role-specific permissions throughout. Done alongside a B.Tech.
- Automated the certification workflow: creating a job card generates a PDF report and emails invitations to the observer and welders; the observer's results generate a certificate for the tester to review; once approved, it is emailed to the welder with a generated welder ID. Admins get an activity log of who did what.
- Deployed and ran the whole system on an AWS EC2 instance, including the VM's firewall configuration and shell scripting for server tasks.

Node.js, MongoDB, PDFKit, EmailJS, AWS EC2, Linux, Shell scripting

Web Developer Intern, Bendra Aerospace Pvt. Ltd.

Trivandrum | May 2022 to Jun 2022

- One of two developers who restructured and redesigned the company's WordPress site to the client's requirements: theme customisation, page restructuring and plugin configuration. The client ultimately chose a simpler design than the one the team proposed.

WordPress

Projects

AURA

May to Jul 2026

An AI coding workspace whose agent works from a live knowledge graph of the project. github.com/abhiram200/AURA

- Browser-based development environment (file explorer, CodeMirror editor, real terminals via xterm.js and node-pty, git tree view) with models running locally through Ollama.
- Agent with about 40 tools that answers from a live code knowledge graph (SYNAPSE) instead of pasting the codebase into context; splits large questions across sub-agents, compacts long conversations and detects repeated tool calls without progress.
- Extended parsing to 17 Tree-sitter grammars. About 48,000 lines of TypeScript; built with Claude Code as an AI-assisted workflow, reviewing specs and designs and steering fixes for hard bugs.

TypeScript, React, Express, WebSockets, node-pty, Tree-sitter, Ollama, Playwright, three.js

SYNAPSE

2026

A code knowledge graph that gives an AI agent near-instant understanding of a project. github.com/abhiram200/SYNAPSE

- Tree-sitter parsing into a graph of files, functions, classes and their imports, calls, extends and implements relationships, re-indexed incrementally by a file watcher.
- Impact analysis with propagation-based severity, hybrid BM25 and embedding search with Reciprocal Rank Fusion, route and API analysis (Express, Flask, FastAPI, Django), git-diff blast radius and dry-run renames.
- About 16,000 lines of TypeScript with 22 HTTP endpoints for projects; inspired by GitNexus, independently implemented; built with an AI-assisted workflow.

TypeScript, Tree-sitter, Express, Server-sent events, React, react-three-fiber, Ollama

Reverse Learning

Aug 2026

An AI learning platform that measures understanding, not completion. reverse-learning.vercel.app/signin

- AI learning platform that measures understanding: learners explain concepts back, and weak understanding inserts remediation and keeps dependent topics gated.
- "AI produces scores, code produces decisions": a pure TypeScript decision engine, isolated by ESLint rules, decides advance, remediate or gate; model output validated with Zod and repaired or retried up to three times.
- Long-running resource discovery moved to a Postgres job queue drained by GitHub Actions, claimed with FOR UPDATE SKIP LOCKED; per-learner inference keys; 59 test files that run without a database or network. Built with an AI-assisted workflow.

Next.js, TypeScript, Prisma, PostgreSQL (Neon), NextAuth (Google sign-in), Zod, Vitest, Vercel, GitHub Actions

ForcingKnight

2026

A neural chess engine, trained from scratch on a rented GPU and shipped as a desktop app.
github.com/abhiram200/chessai

- AlphaZero-style policy-and-value ResNet trained in PyTorch on up to 100 million positions from parallel Stockfish workers and the Lichess evaluations database, streamed out-of-core.
- PUCT MCTS with an exact solver, batched evaluation with virtual loss, and a forcing-line tactical search; replaced ReduceLROnPlateau, which never lowered the learning rate over 100k-step epochs, with a per-step warmup and cosine schedule, and convergence became steady.
- Estimated about 1972 Elo on a Maia (human-scale) ladder and about 1838 on a Stockfish ladder via maximum-likelihood fits (estimates, not official ratings). Shipped as a PySide6 app, UCI engine and Windows build.

Python, PyTorch, python-chess, Stockfish, PySide6, PyInstaller

Vehicle-Based Disaster Alerting System Using LoRa

2024 to 2025

Buried sensors broadcast disaster alerts over LoRa to drivers nearby. ijirt.org/publishedpaper/IJIRT180560_PAPER.pdf

- Team of four: buried Arduino sensor capsules broadcast distress messages over LoRa (ESP32), rebroadcast by relay nodes on poles.
- Wrote the software for the in-vehicle edge unit (Raspberry Pi, LoRa receiver, display) that filters noise with threshold-based checks and alerts drivers. Working prototype; published in IJIRT, June 2025.

Raspberry Pi, LoRa, ESP32, Arduino

Skills

Data engineering: PySpark, Scala, SQL, Python, Databricks, Tidal Enterprise Scheduler, Oracle Cloud Infrastructure

AI and machine learning: PyTorch, LLM agents and tool use, Structured LLM output with Zod, LLM evaluation suites, Hybrid BM25 and embedding search, MCTS, Ollama

Backend and cloud: Node.js, Express, TypeScript, MongoDB, PostgreSQL with Prisma, AWS EC2 (including VM firewall setup), Linux servers, Shell scripting, Vercel, GitHub Actions

Frontend: React, Next.js, Tailwind CSS, three.js, HTML and CSS

AI-assisted development: Directing coding agents (Claude Code) through large codebases by reviewing specs and designs and debugging alongside the agent

Certifications

Databricks Certified Generative AI Engineer Associate
Databricks, Oct 2026 (valid to Oct 2028). Verify

Databricks Certified Data Engineer Associate
Databricks, Apr 2026 (valid to Apr 2028). Verify

GitHub Copilot
Microsoft, Feb 2026 (valid to Feb 2028). Verify

Volunteering

Volunteer, website team lead, TEDxRIET (2023)
Volunteered and led the website team for the college's first approved TEDx event. The site had a speaker list, an event schedule, a registration form and a countdown to the event.

Education

B.Tech in Computer Science and Engineering
Rajadhani Institute of Engineering and Technology, Trivandrum, 2021 to 2025

Publication

"Vehicle-Based Disaster Alerting System Using LoRa Communication", International Journal of Innovative Research in Technology (IJIRT), Vol. 12, Issue 1, June 2025. Abhiram RS, Arjun RM, Pranav J, Muhammad Shafi MS, guided by Ms. Athira Sarath. PDF

Languages: English, Hindi, Malayalam